Indian Institute of Engineering Science and Technology, Shibpur

B.Tech 3rd Semester End-Semester Examination, November 2023

Mathematics-III(MA 2101)

Full Marks : 50 **Time : 3 hours**

Answer any five questions

1. a) The probability that at least one of the events A and B occurs is $\frac{3}{5}$. If A and B occur simultaneously with probability $\frac{1}{5}$, then find $P(\bar{A}) + P(\bar{B})$, where $\bar{A}$ and $\bar{B}$ are the complementary events of A and B respectively.

b) Given for any two events A and B, $P(A) = \frac{1}{4}$, $P(B/A) = \frac{1}{2}$ and $P(A/B) = \frac{1}{4}$; show that A and B are independent.

c) The probability that a man live next 25 years is $\frac{3}{5}$ and that of his wife is $\frac{2}{3}$. Find the probability of that
(i) both will alive (ii) only the man be alive (iii) only the wife will be alive (iv) at least one of them will be alive.

d) Examine whether the function $F(x) = \frac{2}{\pi}tan^{-1}x$, $(-\infty < x < \infty)$ is a distribution function.

[2+2+4+2]

2. a) State and prove Tchebycheff inequality for a continuous random variable.

b) The probability density function of a two-dimensional random variable (X, Y) is given by

$$f(x,y) = \begin{cases} k(x+y), & x \geq 0,\ y \geq 0,\ 0 < x+y < 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find k and evaluate $P(X < \frac{1}{2},\ Y > \frac{1}{4})$.

[(1+5)+(2+2)]

3. a) Show that the correlation coefficient (ρ) between two random variables X and Y, satisfies the relation: $\rho^2 \leq 1$.

b) Fit a straight line to the following data:

x_i	y_i
1	2.4
2	3
3	3.6
4	4
6	5
8	6

[5+5]

P.T.O